# Lecture 03 – Text Functions in Excel

## Module 1: Introduction to Text Functions

Text Functions are used to manipulate, clean, extract, combine, and format text data stored in Excel. Unlike mathematical functions, text functions work with characters, words, and strings.

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### Why are Text Functions Important?

Real-world datasets are rarely clean.
You may encounter:

- Customer names in inconsistent cases.
- Leading or trailing spaces.
- Full names stored in a single column.
- Email IDs requiring username extraction.
- Product codes with prefixes.
- Employee IDs with inconsistent formatting.

Text functions help transform messy data into a structured format suitable for analysis.

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### Business Use Cases

Text Functions are commonly used for:

- Cleaning customer names.
- Extracting first and last names.
- Formatting employee IDs.
- Creating email usernames.
- Standardizing city names.
- Preparing datasets for lookups and reporting.

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### 1: LEFT

The **LEFT** function extracts a specified number of characters from the **left side** of a text string.
**Purpose:** Used when the required information appears at the beginning of a text value.
**Syntax** =LEFT(text, [num_chars])
**Syntax Explanation**

| Argument | Meaning |
|---|---|
| text | Cell containing the text |
| num_chars | Number of characters to extract from the left (Optional, default = 1) |

**Example**
Cell A2
ORD-100001
**Formula** =LEFT(A2,3)
**Result** = ORD

---

**Business Example**
Extract the prefix from Order IDs.

| Order ID | Result |
|---|---|
| ORD-100001 | ORD |
| INV-1025 | INV |
| EMP-250 | EMP |

**Common Uses**

- Invoice Prefix
- Product Category Code
- Employee Code
- Region Prefix

## 2: RIGHT

The **RIGHT** function extracts characters from the **right side** of a text string.
**Purpose:** Used when useful information is located at the end of the text.
**Syntax** =RIGHT(text,[num_chars])
**Syntax Explanation**

| Argument | Meaning |
|---|---|
| text | Text string |
| num_chars | Characters to extract from the right |

**Example**
ORD-100001
**Formula** =RIGHT(A2,6)
**Result**: 100001

**Business Example**
Extract Customer ID from
CUS-25015
**Formula** =RIGHT(A2,5)
**Result**: 25015

**Common Uses**

- Employee Number
- Customer ID
- Invoice Number
- Product Number

## 3: MID

The **MID** function extracts characters from the middle of a text string.
**Purpose:** Used when the required information is neither at the beginning nor at the end.
**Syntax** = MID(text,start_num,num_chars)
**Syntax Explanation**

| Argument | Meaning |
|---|---|
| text | Original text |
| start_num | Starting character position |
| num_chars | Number of characters to extract |

**Example**
ORD-100001
**Formula** = MID(A2,5,6)
**Result**: 100001

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**Business Example**
Extract product code from
PRD-LAP-258
**Formula** = MID(A2,5,3)
**Result:** LAP

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**Common Uses**

- Product Codes
- Customer Codes
- Department Codes
- Employee IDs

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## 4: LEN

The **LEN** function counts the total number of characters in a text string.
It counts:

- Letters
- Numbers
- Spaces
- Symbols

**Purpose:** Used to measure text length and validate data.
**Syntax** = LEN(text)
**Syntax Explanation**

| Argument | Meaning |
|---|---|
| text | Cell or text whose length is to be calculated |

**Example**
Rahul Singh
**Formula** = LEN(A2)
**Result:** 12
(Spaces are counted as characters.)

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**Business Example**
Check whether Customer IDs always contain 10 characters.
Employee ID
EMP100245
**Formula** =LEN(A2)
**Result**: 9

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**Common Uses**

- Validate Phone Numbers

- Check PAN Length
- Verify Aadhaar Digits
- Validate Product Codes
- Password Length Checks

**Summary**

| Function | Purpose | Example |
|---|---|---|
| **LEFT** | Extract characters from the left | =LEFT(A2,3) |
| **RIGHT** | Extract characters from the right | =RIGHT(A2,6) |
| **MID** | Extract characters from the middle | =MID(A2,5,6) |
| **LEN** | Count total characters | =LEN(A2) |

## Module 2: Text Cleaning & Standardization Functions

In real-world business datasets, text data is often inconsistent. Before performing lookups, analysis, or creating dashboards, analysts spend a significant amount of time cleaning and standardizing text.

### 5: TRIM

The **TRIM** function removes **extra spaces** from a text string while keeping only a single space between words.

It removes:

- Leading spaces (before the text)
- Trailing spaces (after the text)
- Multiple spaces between words

**Purpose:** Used to clean text data before analysis or lookups.

**Syntax** = TRIM(text)

**Syntax Explanation**

| Argument | Meaning |
|---|---|
| text | The text or cell reference to clean |

**Example**

Cell A2

Rahul     Singh

Formula

=TRIM(A2)

Result

Rahul Singh

**Business Example**

Customer names imported from an ERP system often contain unnecessary spaces.

Before

Priya      Gupta

After TRIM

Priya Gupta

**Common Uses**

- Cleaning imported datasets
- Preparing data for VLOOKUP/XLOOKUP
- Removing accidental spaces
- Standardizing customer names

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## 6: UPPER

**Definition**

The **UPPER** function converts all lowercase letters into uppercase.

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**Purpose**

Used to standardize text in uppercase format.

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**Syntax**

=UPPER(text)

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**Syntax Explanation**

| Argument | Meaning |
|---|---|
| text | Text to convert into uppercase |

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**Example**

Rahul Singh

**Formula** =UPPER(A2)

**Result**: RAHUL SINGH

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**Business Example**

Convert product codes into uppercase.

Before

lap001

After

LAP001

---

**Common Uses**

- Product Codes
- Employee IDs
- Customer IDs
- Region Codes

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## 7: LOWER

The **LOWER** function converts all uppercase letters into lowercase.

**Purpose:** Used when lowercase text is required.

**Syntax** =LOWER(text)

**Syntax Explanation**

| Argument | Meaning |
|---|---|
| text | Text to convert into lowercase |

**Example**
RAHUL SINGH
**Formula** =LOWER(A2)
**Result**: rahul singh

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**Business Example**
Create email usernames.
Customer Name
Rahul Singh
Result
rahul singh

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**Common Uses**

- Email IDs
- Website URLs
- Login IDs
- Usernames

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## 8: PROPER

The **PROPER** function capitalizes the first letter of every word and converts all remaining letters to lowercase.
**Purpose:** Used to standardize names and titles.
**Syntax** =PROPER(text)
**Syntax Explanation**

| Argument | Meaning |
|---|---|
| text | Text to convert into Proper Case |

**Example**
rAhUl siNGh
**Formula** =PROPER(A2)
**Result**: Rahul Singh

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**Business Example**
Customer names imported from multiple sources may appear in different formats.
Before
ANANYA GUPTA
or
ananya gupta
After PROPER
Ananya Gupta

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**Common Uses**

- Customer Names
- Employee Names
- City Names
- Department Names

- Product Names

**Difference Between UPPER, LOWER and PROPER**

| Function | Output |
|---|---|
| UPPER | RAHUL SINGH |
| LOWER | rahul singh |
| PROPER | Rahul Singh |

**Real Business Scenario**

A retail company receives customer data from different branches.

| Original Data |
|---|
| RAHUL SINGH |
| rahul singh |
| rAhUl SiNgH |
| Rahul Singh |

The company wants all names in a consistent format.

**Formula** =PROPER(TRIM(A2))

This formula:

1. Removes unnecessary spaces using **TRIM**
2. Converts the text into proper case using **PROPER**

**Result**: Rahul Singh

This combination is one of the most commonly used formulas in data cleaning.

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**Best Practices**

- Use **TRIM** before performing lookups to avoid mismatches caused by hidden spaces.
- Use **PROPER** for names, cities, and departments.
- Use **UPPER** for IDs and product codes.
- Use **LOWER** for usernames, URLs, and email addresses.
- Combine multiple text functions when cleaning imported business data.

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**Summary**

| Function | Purpose | Example |
|---|---|---|
| TRIM | Removes extra spaces | =TRIM(A2) |
| UPPER | Converts text to uppercase | =UPPER(A2) |
| LOWER | Converts text to lowercase | =LOWER(A2) |
| PROPER | Capitalizes the first letter of each word | =PROPER(A2) |

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**Interview Questions**

1. What is the difference between **TRIM** and **CLEAN**?
2. Why is **TRIM** commonly used before **VLOOKUP** or **XLOOKUP**?
3. What is the difference between **UPPER**, **LOWER**, and **PROPER**?
4. How would you standardize customer names imported from multiple systems?
5. Write a formula to convert " rahul singh " into "Rahul Singh".

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## Module 3: Search & Replace Text Functions

These functions are widely used by Data Analysts to **locate, extract, replace, and standardize text** in business datasets.

They become especially useful when working with:

- Customer IDs
- Order IDs
- Product Codes
- Email Addresses
- Invoice Numbers
- Employee IDs

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### 9: FIND

- The **FIND** function returns the position of a specific character or text within another text string.
- It is **case-sensitive**, meaning it distinguishes between uppercase and lowercase letters.

**Purpose:** Used to determine the position of characters before extracting data.

**Syntax** =FIND(find_text,within_text,[start_num])

**Syntax Explanation**

| Argument | Meaning |
|---|---|
| **find_text** | Text to locate |
| **within_text** | Text where the search is performed |
| **start_num** | Starting position (Optional) |

**Example**

Order ID

ORD-100001

**Formula** =FIND("-",A2)

**Result**: 4

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**Business Example**

Find the position of "-" before extracting Order Number.

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**Common Uses**

- Locate separators
- Extract IDs
- Email username extraction
- Product Code extraction

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## 10: SEARCH

- The **SEARCH** function returns the position of text inside another text string.
- Unlike FIND, **SEARCH is NOT case-sensitive.**

**Purpose:** Used when case sensitivity is not required.

**Syntax** =SEARCH(find_text,within_text,[start_num])

**Syntax Explanation**

| Argument | Meaning |
|---|---|
| **find_text** | Text to search |
| **within_text** | Text containing the search value |
| **start_num** | Starting position (Optional) |

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**Example**

Rahul Singh

**Formula** =SEARCH("singh",A2)

**Result**: 7

Even if written as:

SINGH

or

Singh

SEARCH still returns the correct position.

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**Business Example**

Locate "@"

rahul@gmail.com

**Formula** =SEARCH("@",A2)

**Result**: 6

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**Difference Between FIND and SEARCH**

| FIND | SEARCH |
|---|---|
| **Case-sensitive** | Not case-sensitive |
| **Exact character matching** | Flexible matching |
| **Faster** | Slightly slower |
| **Best for IDs and Codes** | Best for names and emails |

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## 11: REPLACE

The **REPLACE** function replaces characters based on their position.

Instead of searching for text, it replaces characters starting from a specified position.

**Purpose:** Used when character positions are known.

**Syntax** =REPLACE(old_text,start_num,num_chars,new_text)

**Syntax Explanation**

| Argument | Meaning |
|---|---|
| **old_text** | Original text |
| **start_num** | Starting position |
| **num_chars** | Number of characters to replace |

| | |
|---|---|
| **new_text** | Replacement text |

**Example**

ORD-100001

**Formula** =REPLACE(A2,1,3,"INV")

**Result**: INV-100001

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**Business Example**

Convert

EMP-101

to

EMPLOYEE-101

**Formula** =REPLACE(A2,1,3,"EMPLOYEE")

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**Common Uses**

- Update prefixes
- Change department codes
- Standardize invoice formats

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## 12: SUBSTITUTE

- The **SUBSTITUTE** function replaces specific text with another text.
- Unlike REPLACE, it replaces based on the actual text rather than character position.

**Purpose:** Used when the exact text is known.

**Syntax** =SUBSTITUTE(text,old_text,new_text,[instance_num])

**Syntax Explanation**

| Argument | Meaning |
|---|---|
| **text** | Original text |
| **old_text** | Text to replace |
| **new_text** | New text |
| **instance_num** | Which occurrence to replace (Optional) |

**Example**

Laptop Computer

**Formula** =SUBSTITUTE(A2,"Laptop","Notebook")

**Result**: Notebook Computer

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**Business Example**

Replace city names.

**Before**

Bombay

**After**

Mumbai

**Formula** =SUBSTITUTE(A2,"Bombay","Mumbai")

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**Another Example**

Replace only the first occurrence.

Apple Apple Apple

**Formula** =SUBSTITUTE(A2,"Apple","Orange",1)
**Result**: Orange Apple Apple

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**Difference Between REPLACE and SUBSTITUTE**

| REPLACE | SUBSTITUTE |
|---|---|
| **Replaces by character position** | Replaces by matching text |
| **Position must be known** | Text must be known |
| **Useful for IDs** | Useful for names and words |

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**Real Business Scenario**
A company stores Order IDs as:
ORD-100001
ORD-100002
ORD-100003
The ERP system now requires all Order IDs to begin with **INV**.
**Formula** =REPLACE(A2,1,3,"INV")
**Result**: INV-100001

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The company also changes city names.
**Before**
Bombay
**After**
Mumbai
**Formula** =SUBSTITUTE(A2,"Bombay","Mumbai")

---

**Best Practices**

- Use **FIND** when case sensitivity matters.
- Use **SEARCH** for flexible searches regardless of letter case.
- Use **REPLACE** when you know the character positions.
- Use **SUBSTITUTE** when replacing specific words or text values.
- Combine these functions with **LEFT**, **RIGHT**, and **MID** for advanced text extraction.

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**Summary**

| Function | Purpose | Example |
|---|---|---|
| FIND | Finds position (case-sensitive) | =FIND("-",A2) |
| SEARCH | Finds position (not case-sensitive) | =SEARCH("@",A2) |
| REPLACE | Replaces characters by position | =REPLACE(A2,1,3,"INV") |
| SUBSTITUTE | Replaces matching text | =SUBSTITUTE(A2,"Bombay","Mumbai") |

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**Interview Questions**
**Q1. What is the difference between FIND and SEARCH?**
**Hint:** Think about case sensitivity.

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**Q2. What is the difference between REPLACE and SUBSTITUTE?**
**Hint:** One works with positions, the other with matching text.

**Q3. Which function would you use to replace all occurrences of "Delhi" with "New Delhi"?**
**Hint:** Text-based replacement.

**Q4. How would you extract the Order Number from ORD-100001?**
**Hint:** Combine **FIND** with **MID** or **RIGHT**.

**Q5. A customer email contains "rahul.singh@gmail.com". Which function would you use to find the position of "@" before extracting the username?**
**Hint:** Search for a specific character.

## Module 4: Text Combining & Conversion Functions

In business reporting, data is often stored across multiple columns. For example:

| First Name | Last Name |
|---|---|
| Rahul | Singh |

or

| City | State |
|---|---|
| Lucknow | Uttar Pradesh |

To prepare reports, dashboards, or exports, these values often need to be combined into a single cell.
Similarly, imported data may store numbers as text, requiring conversion before calculations.
These functions solve those problems.

### 13: CONCAT

- The **CONCAT** function combines two or more text strings into a single text string.
- It is the modern replacement for **CONCATENATE**.

**Purpose:** Used to merge multiple cells or text values into one.
**Syntax** =CONCAT(text1,[text2],...)
**Syntax Explanation**

| Argument | Meaning |
|---|---|
| text1 | First text or cell reference |
| text2 | Second text or cell reference (Optional) |
| ... | Additional text values |

**Example**

| A | B |
|---|---|
| Rahul | Singh |

**Formula** =CONCAT(A2,B2)
**Result**: RahulSingh

**Add Space Between Names**
**Formula** =CONCAT(A2," ",B2)
**Result**: Rahul Singh

**Business Example**

Create Full Customer Name

| First Name | Last Name |
|---|---|
| **Priya** | Gupta |

**Formula** =CONCAT(A2," ",B2)

**Result**: Priya Gupta

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**Common Uses**

- Full Name
- Complete Address
- Employee Name
- Product Description
- Customer Code

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## 14: TEXTJOIN

- The **TEXTJOIN** function combines multiple text values using a specified delimiter.
- Unlike CONCAT, you only specify the separator once.

**Purpose:** Used to combine text with spaces, commas, hyphens, or other separators.

**Syntax** =TEXTJOIN(delimiter,ignore_empty,text1,[text2],...)

**Syntax Explanation**

| Argument | Meaning |
|---|---|
| **delimiter** | Character placed between text values |
| **ignore_empty** | TRUE ignores blank cells, FALSE includes them |
| **text1** | First text |
| **text2** | Additional text |

**Example**

| First | Middle | Last |
|---|---|---|
| **Rahul** | Kumar | Singh |

**Formula** =TEXTJOIN(" ",TRUE,A2:C2)

**Result**: Rahul Kumar Singh

---

**Business Example**

Combine Address

| City | State | Country |
|---|---|---|
| **Lucknow** | Uttar Pradesh | India |

**Formula** =TEXTJOIN(", ",TRUE,A2:C2)

**Result**: Lucknow, Uttar Pradesh, India

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**Advantages over CONCAT**

- Adds separators automatically.
- Ignores blank cells.
- Cleaner formulas.
- Easier to manage.

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**Common Uses**

- Address Creation

- Employee Full Name
- Report Titles
- Customer Details

**Difference Between CONCAT and TEXTJOIN**

| CONCAT | TEXTJOIN |
|---|---|
| **No automatic separator** | Supports delimiter |
| **Manual spaces required** | Delimiter specified once |
| **Doesn't ignore blanks** | Can ignore blank cells |
| **Simpler** | More flexible |

## 15: TEXT

The **TEXT** function converts a number, date, or time into text using a specified format.
**Purpose:** Used to display values in a customized format without changing the original value.
**Syntax** =TEXT(value,format_text)
**Syntax Explanation**

| Argument | Meaning |
|---|---|
| **value** | Number, date or time |
| **format_text** | Desired display format |

**Example 1**
Cell
45000
**Formula** =TEXT(A2,"₹#,##0")
**Result**: ₹45,000
**Example 2**
Date
01/07/2026
**Formula** =TEXT(A2,"dd-mmm-yyyy")
**Result**: 01-Jul-2026
**Example 3**
**Formula** =TEXT(A2,"dddd")
**Result:** Wednesday

**Business Uses**
- Dashboard Titles
- Invoice Dates
- Financial Reports
- Custom Number Formats

## 16: VALUE

The **VALUE** function converts numbers stored as text into actual numeric values.
**Purpose:** Used when imported datasets contain numbers stored as text.
**Syntax** =VALUE(text)
**Syntax Explanation**

| Argument | Meaning |
|---|---|
| text | Text representing a numeric value |

**Example**

Cell

"2500"

**Formula** =VALUE(A2)

**Result**: 2500

Now Excel recognizes it as a number.

---

**Business Example**

An ERP export stores sales values as text.

**Before**

"15000"

Excel cannot calculate totals correctly.

**Formula** =VALUE(A2)

**Result**: 15000

Now SUM, AVERAGE, and other mathematical functions work correctly.

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**Common Uses**

- Imported CSV files
- ERP Exports
- Banking Data
- Financial Reports
- Data Cleaning

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**Real Business Scenario**

A company stores customer information in separate columns.

| First | Last | City |
|---|---|---|
| Rahul | Singh | Delhi |

To generate a report:

=TEXTJOIN(" - ",TRUE,A2:C2)

**Result**: Rahul - Singh - Delhi

---

The finance team imports revenue data:

"250000"

Instead of

250000

**Formula** =VALUE(A2)

Now Excel can calculate:

- SUM
- AVERAGE
- MIN
- MAX
- Pivot Tables

without any issues.

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**Best Practices**

- Use **CONCAT** for simple text combinations.
- Use **TEXTJOIN** when separators are required or blank cells should be ignored.
- Use **TEXT** to display numbers and dates in professional report formats.
- Use **VALUE** before performing calculations on imported numeric text.

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**Summary**

| Function | Purpose | Example |
|---|---|---|
| **CONCAT** | Combines text | =CONCAT(A2," ",B2) |
| **TEXTJOIN** | Combines text with delimiters | =TEXTJOIN(", ",TRUE,A2:C2) |
| **TEXT** | Formats numbers, dates, and times as text | =TEXT(A2,"dd-mmm-yyyy") |
| **VALUE** | Converts text to numbers | =VALUE(A2) |

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**Interview Questions**

**Q1. What is the difference between CONCAT and TEXTJOIN?**
**Hint:** Think about delimiters and blank cells.

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**Q2. Why would you use the TEXT function instead of changing the cell format?**
**Hint:** Consider dynamic report generation.

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**Q3. A CSV file imports sales values as text. Which function would you use before applying SUM or AVERAGE?**
**Hint:** Convert text to numbers.

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**Q4. Write a formula to combine First Name, Last Name, and City separated by commas.**
**Hint:** Use TEXTJOIN.

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**Q5. Why do Data Analysts frequently use the VALUE function when working with ERP or CSV exports?**
**Hint:** Think about calculations on imported data.

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